

# DISTANCE MEASURING DEVICE

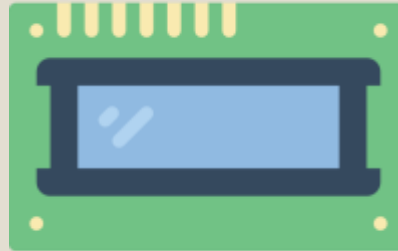
(USING ULTRASONIC SENSOR & ARDUINO UNO)

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# REQUIRED MATERIALS



**Arduino uno**

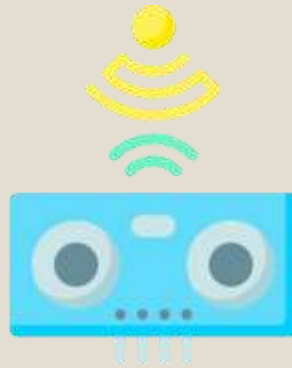


**16x2 LCD Display**

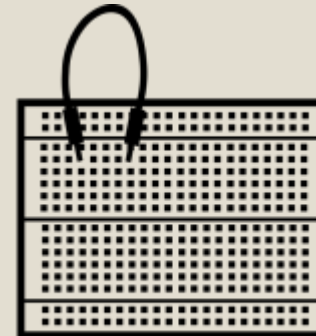


Created by Duvi  
from Noun Project

**Jumper wires**



**Ultrasonic Sensor**



Created by Stephanie Piper  
from Noun Project

**Breadboard**

# COMPONENT EXPLANATION

## ➤ ARDUINO UNO

Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button.

# COMPONENT EXPLANATION

## ➤ ULTRASONIC SENSOR

An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. An ultrasonic sensor uses a transducer to send and receive ultrasonic pulses that relay back information about an object's proximity.

# COMPONENT EXPLANATION

## ➤ LCD DISPLAY WITH I2C MODULE

I2C LCD requires only 2 lines (SDA and SCL) for transferring the data. So, the complexity of the circuit is reduced. Interfacing I2C LCD to the Arduino: I2C LCD can be connected to the Arduino directly with SDA pin to SDA pin and SCL pin to SCL pin.

# COMPONENT EXPLANATION

## ➤ BREADBOARD

A breadboard can be divided into two segments, which are called the bus strip and the terminal strip. The bus strip consists of two long lines of spring clips running across the board, these lines can be used to provide supply voltage (VCC) and ground (GND) to the circuit.

# COMPONENT EXPLANATION

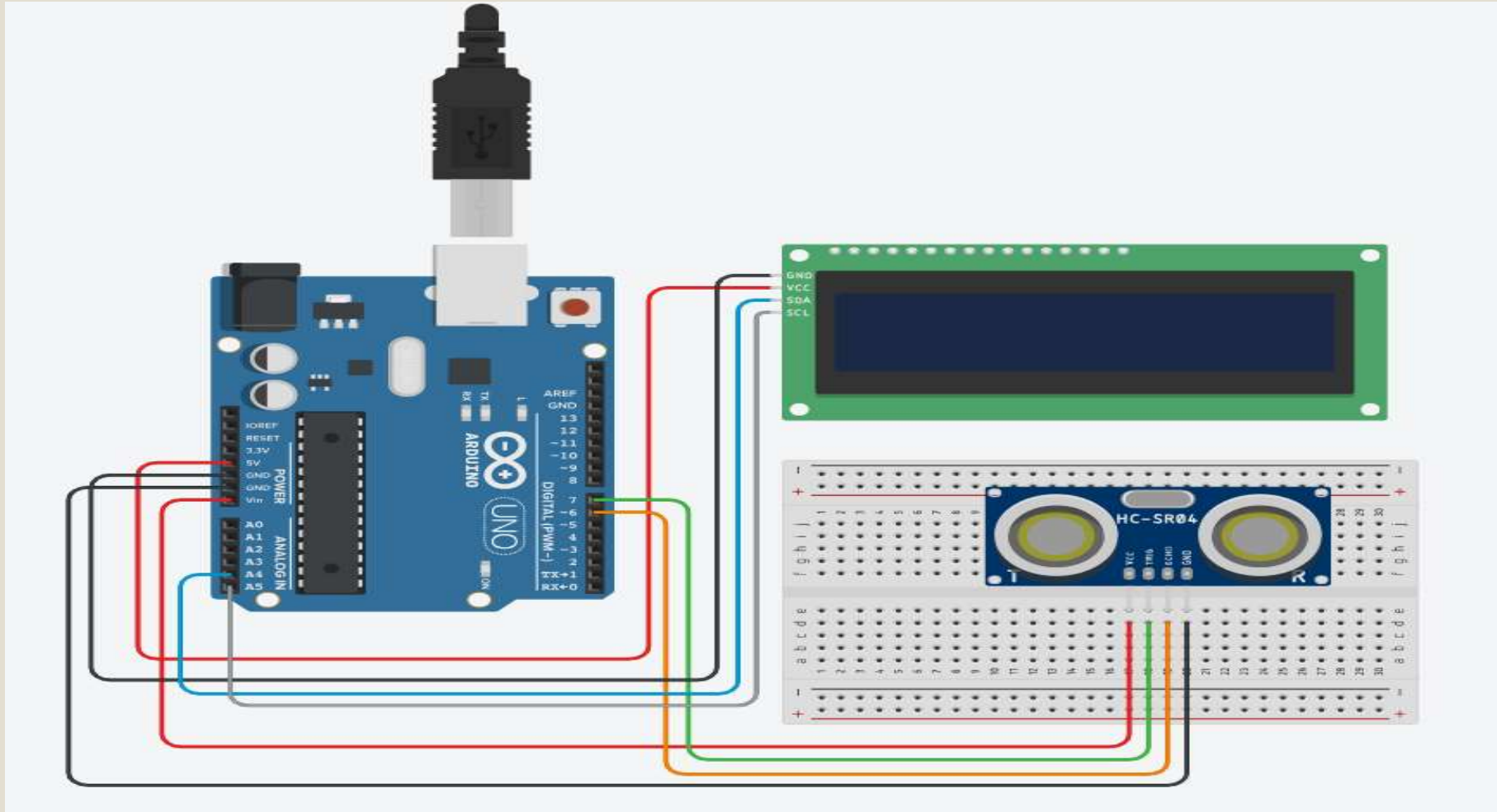
## ➤ JUMPER WIRES

Jumper wires are electrical wires with connector pins at each end. They are used to connect two points in a circuit without soldering. You can use jumper wires to modify a circuit or diagnose problems in a circuit.

# DESCRIPTION

- Distance measuring device project is design to implement an accurate and efficient system of measuring distance in real time.
- The project employed Ultrasonic sensor for distance measuring, utilizing the time-of-flight principle. A circuit board(e.g. Arduino uno which have ATmega 328P microcontroller) processed the sensor data.

# DIAGRAM



# CODE

MINI\_PROJECT.ino

```
1  #include <Wire.h>
2  #include <LiquidCrystal_I2C.h>
3
4  LiquidCrystal_I2C lcd(0x27, 16, 2);
5  #define trigPin 7
6  #define echoPin 6
7
8  void setup()
9  {
10     pinMode(trigPin, OUTPUT);
11     pinMode(echoPin, INPUT);
12     lcd.begin();
13     lcd.backlight();
14     lcd.clear();
15     lcd.setCursor(4,0);
16     lcd.print("TESTING");
17     delay(2000);
18     lcd.setCursor(0,0);
19     lcd.print("Target Distance:");
20 }
21
22 void loop()
23 {
24     long duration, distance;
25     digitalWrite(trigPin, LOW);
```

# CODE

MINI\_PROJECT.ino

```
15  lcd.setCursor(4,0);
16  lcd.print("TESTING");
17  delay(2000);
18  lcd.setCursor(0,0);
19  lcd.print("Target Distance:");
20  }
21
22  void loop()
23  {
24      long duration, distance;
25      digitalWrite(trigPin, LOW);
26      delayMicroseconds(2);
27      digitalWrite(trigPin, HIGH);
28      delayMicroseconds(10);
29      digitalWrite(trigPin, LOW);
30      duration = pulseIn(echoPin, HIGH);
31      distance = (duration/2) / 29.1;
32
33      lcd.setCursor(0,1);
34      lcd.print("                ");
35      lcd.setCursor(0,1);
36      lcd.print(distance);
37      lcd.print(" cm");
38      delay(250);
39  }
```

Output